

A Complex Human-Machine Coordination Problem: Essential Constraints on Interaction Control in Bionic Communication Systems

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ABSTRACT

Novel voice enhancements for larynx amputees utilize a variety of wearable interfaces and/or artificial intelligence. However, the problem formulation remains posed as “voice restoration”, which stems from common, but limiting, assumptions about communication as a reliable signal transfer, in which “signal” equals to voice. Yet beyond this signal transfer, the vital part of communication is mutual influence. During conversations people coordinate their movement dynamics, i.e. adjust speech timing, intonation and gestures. By taking this perspective, informed by the newest trends in the cognitive sciences (ecological, enacted, extended cognition) as well as recent studies on human-machine integration, we seek to shed new light on the processes of analysis and evaluation of specific systems for laryngectomees. We propose the integrated framework which enables us to interpret bionic communicative interfaces as parts of distributed cognitive systems that allow for different level of control over interactions. This paper analyses 5 bionic systems, consisting of a laryngectomee supported with i) voice copy, ii) voice conversion, iii) silent speech iv) bionic voice or v) lean-AI interfaces. We demonstrate how using the broader perspective leads to noticing important aspects of communication and informs a more human-centered design and evaluation.

CCS CONCEPTS •Human-centered computing → Human computer interaction (HCI) → HCI theory, concepts and models •Human-centered computing → Interaction design → Interaction design theory, concepts and paradigms

KEYWORDS

Laryngectomy, Bionic systems, Coordination, Communication, Human-machine Interaction, Voice Copy, Voice Conversion, Silent Speech Interfaces, Bionic Voices, Lean-AI

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1 INTRODUCTION

There is a variety of novel devices supporting communication of people who have lost their larynx. Many solutions utilize wearable interfaces, which may be enriched by AI or signal processing computations. As natural conversation requires ongoing adaptation to particular situations and interlocutors, the question arises about the needs and kinds of control over vocal characteristics, which has to be exerted to enable fluent interaction. This paper presents a theoretical framework, with an aim to provoke discussions on in-depth analyses of laryngectomees' interactions mediated via supporting devices.

Firstly, we present the anatomy altered after the larynx resection (1.1) and the established understanding of the problem as a signal transfer disturbance (1.2). Then, in Section 2, we introduce the current perspective to cognition as ecological (2.1), enacted (2.2) and extended (2.3) enabling a richer picture of human communicative interaction, in which the goal is to establish and maintain interpersonal relations and exert mutual influence. Next, we note the related concerns in the work on human-computer interaction which is geared towards explaining interactions emergent between humans and machines in terms of complex systems (2.4). We propose that a change of the metaphor for communication from signal transfer (dubbed “conduit” in the literature) to coordination accommodates all of the aforementioned concerns (2.5). On this theoretical background, in Section 3 we analyze the underlying assumptions and functioning of bionic systems, namely laryngectomees supported with: voice copy (3.1), voice conversion (3.2) silent speech (3.3), bionic voice (3.4) or lean-AI (3.5) interfaces. We demonstrate that signal transfer is but one of the elements relevant for communication in these analyses, and that the changed perspective brings forward novel aspects and criteria which may have a crucial role for designing and evaluating these systems. We argue that applying interaction-based measures to the device design and evaluation has the potential to enrich the users' ability to fully engage in social interactions and establish conceptual frames for informative comparisons between different types of systems.

1.1 Laryngectomy: a biophysical perspective

Laryngeal cancer treatment may lead to the amputation of the larynx, an organ involved in voice source production [24, 29]. After the larynx is resected, the air no longer traverses through the mouth and nose, and a new breathing opening (stoma) is created (Figure 1). The redirection of air, as well as the lack of vibrating vocal folds, changes the way of breathing, eating and speaking.

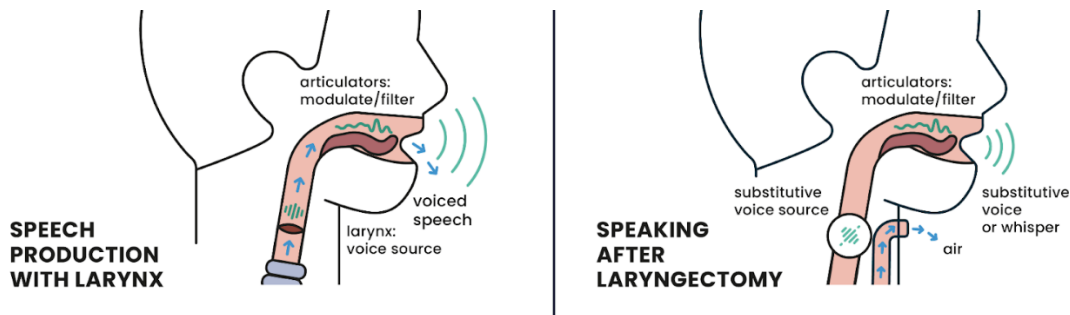


Figure 1: Speech production before and after laryngectomy. The air is redirected to the opening in the neck (stoma). The full voicing speech is possible only when provided with a substitutive voice source. Whispering is possible from the air located in the mouth.

Vibrating vocal folds produce the fundamental source of the voice that allows for intonation modulation. Although they are missing after the surgery, the mouth, tongue and other articulators remain almost intact, allowing laryngectomees to modulate the speech waveform when provided with a sound source (in some other, substitutive, way). The consequences of such a modification are traditionally interpreted as well-defined biological, physical or acoustic problems. The established restorative ways of producing basic vibration of the vocal tract include esophageal speech, voice prosthesis, electronic larynx [29] and a number of novel computing devices and interfaces that have been also introduced recently [12, 13, 19, 20, 24] to compensate for the missing larynx and allow for voiced speech production.

1.2 Simplified problem: “voice restoration”, “information transfer” (conduit metaphor)

The works on laryngectomy tend to describe the condition of laryngectomees as “voice loss” and therefore define the problem for technological solutions to solve as “voice restoration”. Such formulation is based on the view of communication as signal transfer, a simplification termed “conduit metaphor” in cognitive science [45]. The core of communication, as introduced by Shannon in 1948, is the message sent from speaker to listener via a noisy channel [26]. According to the conduit metaphor, the message consists of encoded thoughts which are then deciphered by the listener. Shannon himself used this definition in the context of signal transmission optimization in telecommunication, consciously bracketing it from semantic and pragmatic considerations. Nevertheless, the metaphor has been biasing the way communication was understood in psychology and cognitive science [25] for decades. Verbal communication was thus simplified to signal transfer from one place to another, while other factors, crucial for coordination, were neglected.

2 THEORETICAL CONTRIBUTIONS: INTEGRATING EMBODIED COGNITIVE SCIENCE WITH HCI

The aforementioned view of communication is very far from the one studied over decades within anthropology, biosemiotics, ecological psychology and enactivism. This section briefly presents

the most important contributions of those complementary perspectives, which take into account environment and bodily interaction as crucial for human cognition: the embodied, ecological, enactive, extended cognitive science. We also integrate them with human-computer interaction research, aiming for a comprehensive framework.

2.1 Ecological perspective: language as a system of constraints nested in the field of values

We base our analyses on the conceptual frames rooted in the long-standing traditions of ecological psychology, which appreciates the role of the environment in shaping cognitive processes. Since the 1960s [5, 6] ecological psychologists have claimed that human cognition primarily serves action and should be investigated with respect to the particular environment (physical and social), which provides affordances, possibilities for actions [14, 15]. This perspective broadens the scope of bionic communication analysis to include: 1) a person without a larynx, as an agent realizing certain goals and values, 2) the computing systems they are using, as providing specific affordances 3) their interlocutors in particular situations, who are also providers and users of those affordances, 4) particularly structured environment in which actions are performed. In short, the ecological approach in this context means that performance of the interacting dyad situated in a particular environment comes forward as the unit of analysis.

In the recent works such as [15, 25] where this perspective to language functioning in real life interactions is applied, speech is treated as a crucial means for structuring social interaction but is only one of many components contributing to the coordinative success. It is claimed that the way to go beyond a mechanistic view on communication is to treat utterances as falling on the background of rich interactive dynamics, constraining it to enable fluent values realization. Speech signals are among those constraints, but so are the speech-mediating devices, which are a source of additional constraints.

2.2 Enactive perspective: participatory sense-making

Action-based cognition is at the core of another important framework that has recently been gaining importance in cognitive science [31]. Enactivism points to the active role of interlocutors in creating social situations by their – mutually influencing – behaviors. This approach stresses the importance of first-person experience, and the agency (individuation, the need to be autonomous) is considered to be the organisms' driving force. Consequently, communication is viewed as meaning co-construction or "participatory sense-making", which is dialogical and relies on physical coordination and engaged contribution [10, 11]. Constant adaptation to the interlocutors' bodily dynamics, including those who use artificial systems to support communication, arises from concrete physical interaction that is happening at the particular moment. The first-person experience becomes an important and rich source of data to be used in the analyses of embodied and situated behavior, thus significantly broadening the available methodology.

2.3 Extended perspective: distributed cognitive systems

A complementary approach useful in the present research problem is postulated by the extended mind theory [7, 16, 18]. Its proponents argue that physical devices created by cognitive agents in order to aid functioning are the constitutive parts of the cognitive systems. They allow us to "offload" our cognitive processes to physical and computational architectures (such as calculators and smartphones). Systems supporting a laryngectomee perform certain aspects of speech processing which were, prior to the operation, internal to the given person. When some parts of the speech production process are no longer located in the body of the user, the interplay within the bionic system itself (between the biological part: human, and the technological system: computer) changes the structure of control that is required for coordination in social interactions. Since the computing device performs a part of the work crucial for social interaction, close integration is needed between the laryngectomee and the artificial system. What enables fluent communication (coordination with other people) is the coordination between human and the supportive systems.

2.4 HCI perspective: human-centered design, user experience research

Although historically the presented theoretical grounds for embodied human cognition have been rarely connected explicitly with the applied human-computer interaction research, the latter seems to have developed similar ideas in practice: designing everyday utilities that afford action [23], and fit within the given context and real-life scenario [24]. The concept of agency, that is, the control over one's action, is investigated in depth in case of prosthetics use [9], while disability studies start to notice the first-person perspective of users with compromised speech in the context of voice personal assistants [37]. More recently, the integratory work between enactivism and HCI is being done [38], pointing out to the observed complexity of interactions calling for a novel explanatory

framework. The complex systems theory (important in the ecological perspective as well) is a basis for such integration [39], and the analysis of the continuous and proactive role of AI, e.g. in smart notifications [40]. Some authors take an even further step and call for an integration, not just interaction between the human and the machine [41]. We claim that integrated theoretical perspectives have a potential to provide novel guidelines consistent with the applied work.

2.5 Proposed framework: Communication as mutual influence (coordination metaphor)

The integration of the above theoretical and applied developments allows for a principled novel look at the communicative situation and how it is perturbed after larynx amputation. From this new perspective, human interaction is viewed as a dynamical system in which interlocutors provide mutual constraints to coordinate for diverse tasks, while taking advantage of properties provided by the artificial systems. Instead of a simple "signal transmission" problem, we now have a coordination problem, in which signal transmission is just an element. In the coordination problem, the interacting dyad is the unit of analysis, and notions such as mutual constraints and control come to the fore. While setting the problem as a transmission problem largely dictated the concentration of research and design efforts on the signal itself and its intelligibility to a listener, the coordination problem also includes concerns such as continuous physical coordination of bodies in interaction, first-person feeling of control and issues of agency in interaction with the device and mutual influence with the interlocutor. This perspective allows for analyzing features of devices as enabling or disabling constraints for fluent value-realizing in interactions (Figure 2).

3 BIONIC INTERFACES ANALYSIS IN THE INTEGRATED FRAMEWORK

A variety of solutions emerged over the past decade with the goal to enhance the communication of larynx amputees. In this section we analyze the concrete examples of bionic systems composed of an alaryngeal person supported by a device, conversing with an interlocutor. The aim of those examples is to demonstrate how our conceptual frames bring forth novel important aspects of the problem. We will focus on the following novel communicative devices based on: i) voice copy, ii) voice conversion, iii) silent speech interface, iv) bionic voice or v) lean-AI interface illustrated in sections 3.1-3.5.

In each case we will specify 1) how the system works, 2) what are the essential constraints from the perspective of interaction analysis and interface design, 3) what are the embedded computing models and constraints emerging from them, 4) how the interplay of constraints enables or disables appropriate control. Our aim is to provide the readers with an observation lens allowing for recognition of communication structure, in order to understand better where exactly and which type of control is needed to coordinate, and which design enables/disables such control.

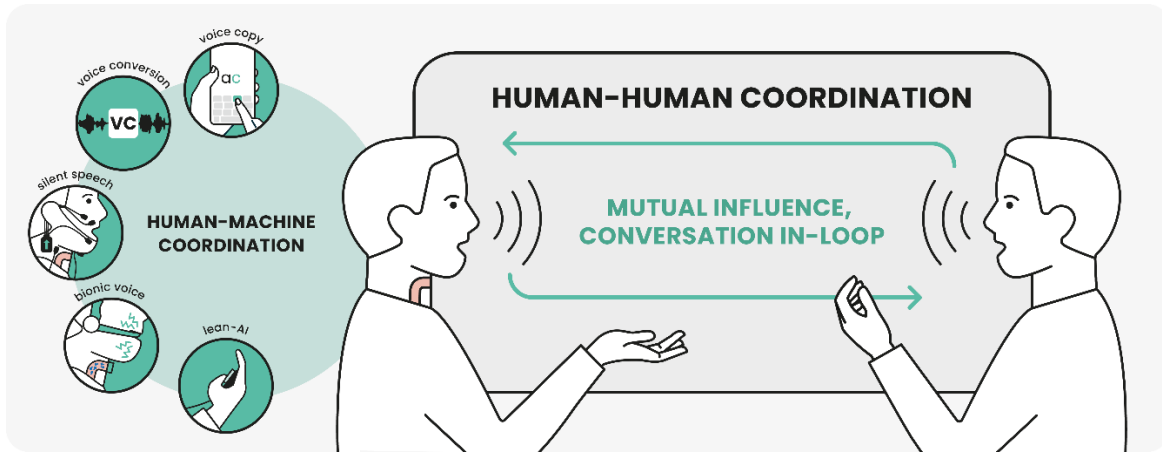


Figure 2: Coordination metaphor: a person in human-machine interaction nested within the human-human interaction.

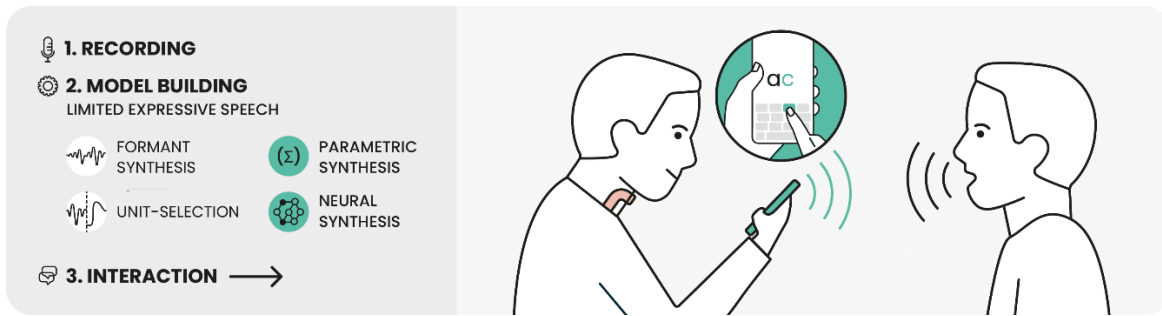


Figure 3: Voice copy training phase: recording of patients’ voice, and model building according to the synthesis technique e.g. neural synthesis. The interaction is mediated by a device for: input text, synthesis and output voice production.

3.1 Personalized text-to-speech system (digital voice copy): retaining “static” personality

Digital voice copy enables a person to embed their own speech characteristics in a text-to-speech system. In a typical scenario, the user records a corpus of 500-1000 sentences, which is then used to build a speech model [24]. The parametric speech models (based on statistical features of the provided speech recordings) or models based on deep neural networks are already used in commercially available systems [32, 33]. This solution has been tested on populations of laryngectomees recording the voice before the surgery and “speaking” with their voice copy afterward [24]. Although the system “remembers” the vocal characteristics of a given person remarkably well, and the “message” is transferred adequately from the signal-transfer point of view, the expressive speech is significantly limited – the voice sounds the same each time the same text is typed on the device. As argued in [3], the conversational speech with the use of TTS also lacks the crucial means for controlling social interaction: disfluencies, laughter, filler pauses etc. The system does not “perform” [4] along with the speaker, as it cannot adjust its vocal features to the situation at hand.

From the point of view of coordination in an interactive situation, other crucial interactive constraints come to the fore: 1) introduction of unnatural devices mediating the interaction: text has to

be typed in and computations performed on a smartphone or a computer (Figure 3) forcing the person using it to look at the screen during the interaction, and thus interrupting the gaze dynamics; 2) obviously the interaction dynamics is significantly changed as the typing and computation needed for conversion introduce significant delays. From the point of view of the conduit metaphor, the voice quality and naturalness will be appreciated in such a system. From the broader perspective, we will note how such “naturalness” also carries the ability to retain the personal characteristics of a particular laryngectomee (first-person feeling of voice ownership). Yet the part of this feeling is the ability to control the interaction with the voice in a certain way, and how using the device changes this embodied interaction dynamics is a part of the problem. This is a source of additional criteria for evaluation of the system according to the type of interaction, task, environment.

3.2 Voice-to-voice conversion: dynamically morphing the speech signal

Another approach to the issue of improving vocal communication is voice conversion, which enhances the quality of voice produced by the laryngectomee (e.g. electrolaryngeal voice or whisper) [46]. The system captures the voice produced by a person, modifies the characteristics of the speech signal to how it was uttered by the

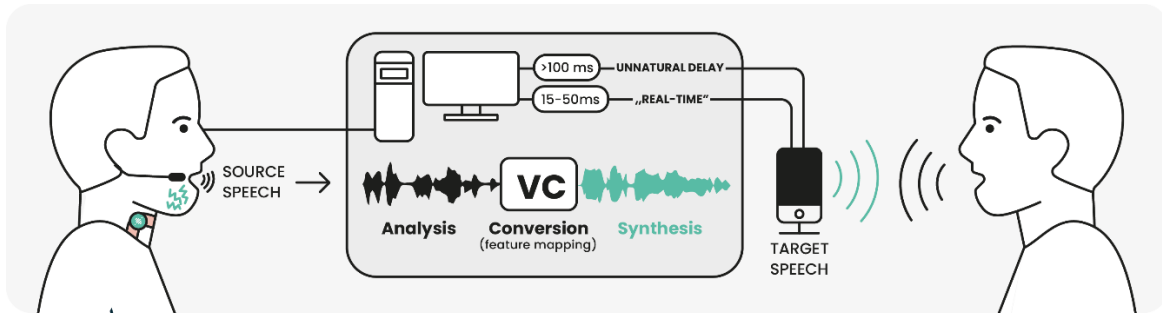


Figure 4: Voice conversion systems alter the speech signal. The linguistic content remains the same, while the vocal characteristics change according to a given transformation function learnt from e.g. parallel corpora of source and target speaker recordings.

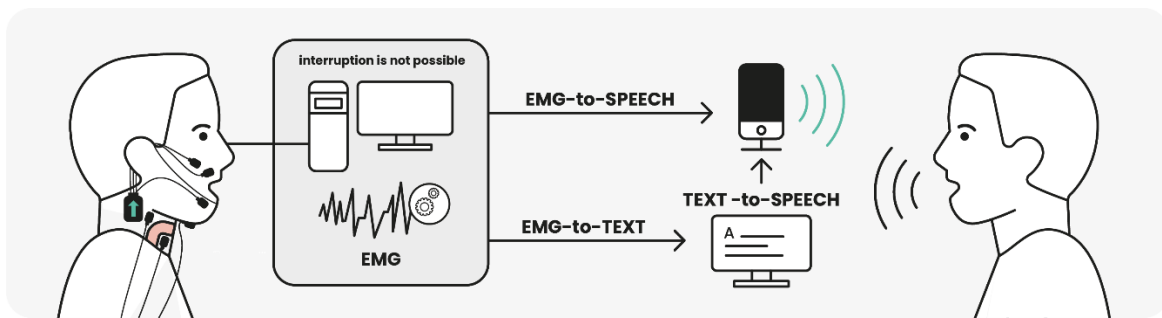


Figure 5: Silent speech interfaces read biological signals e.g. via EMG sensors placed on the face and neck muscles. Then the computing system predicts intended words as text or speech signals directly. The control of the voice is limited, the computer predicts the output

target speaker, and produces the output voice [27]. The interaction with such a system requires a microphone to capture the audio signal from the mouth, the output loudspeaker, and an efficient computing device in between (Figure 4).

Most importantly, the computer system only “smoothes” the sound of the voice, while the actual speech production is on the side of the actual user who has learnt the use of one of the voice-producing means (see 1.1). Such smoothing takes time and presently the designers’ effort is geared towards bringing it down to under 100ms, which arguably preserves the naturalness of speech flow [27]. Such timing has recently become possible utilizing “fully neural” interfaces based on cyclic variational autoencoders and recurrent neural networks [30]. The intelligibility of the transmitted signal relies here on the quality of the morphing model. The coordination in interaction is, however, influenced crucially by the delay.

The crucial outcome of applying our broader framework is noticing: 1) the necessity of locating control over linguistic content and intonation (source speech signal) on the side of the user, 2) additional devices (microphone, speaker) as parts of interacting system, 3) interaction dynamics as an important factor and various timing issues as disturbing its flow.

3.3 Silent speech interfaces: AI controls the crucial moments of interactions

Another solution which has recently been proposed to enhance laryngectomees communication abilities are silent speech interfaces (Figure 5). The biosignals correlated with speech production (such as neuromuscular signals from the face and neck, or directly from the articulators, captured e.g. with EMG sensors) are collected and used by a computational model to predict the intended utterances. Currently two main approaches to that problem are investigated: 1) EMG-to-text combined with a voice copy [34], 2) EMG-to-Speech (direct speech synthesis) [19]. The influence of this solution on coordination in conversation are as follows: 1) devices are located on the mouth of the user, which alters the interaction schemas, 2) the timing of the system’s performance is critical. Out of the two of the above solutions, the direct synthesis is faster and, as claimed by [13], ensures performance to be perceived as real-time and therefore an easy adaptation of the user, allowing for dynamic coordination.

However, the critical difference is conceptual: the final decision on which sentence/phrase/words to produce is left to the artificial system. The human cannot correct the output voice according to the situation at hand using the available contextual information such as the interlocutor’s facial expression or gestures. For example, if a system fails to predict the right utterance, or the user changes their

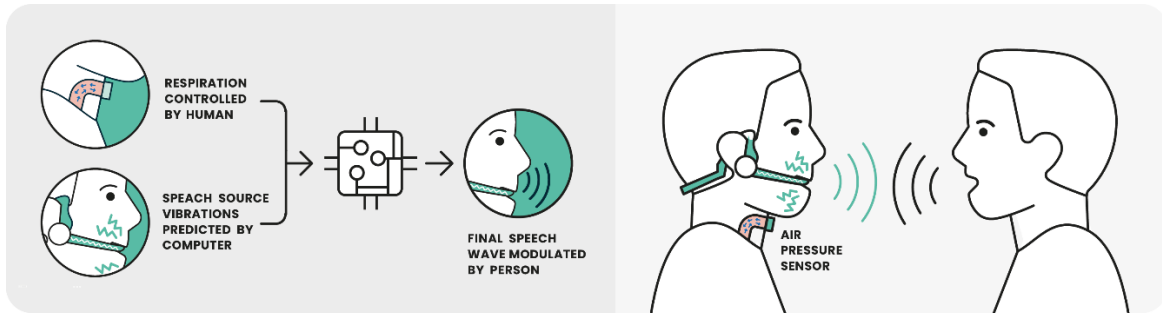


Figure 6: In bionic voice systems, the system captures the signals emitted by the speaker, while the computing device is responsible for predicting the waveshape. The close collaboration between the human and machine is required for a smooth operation of the system.

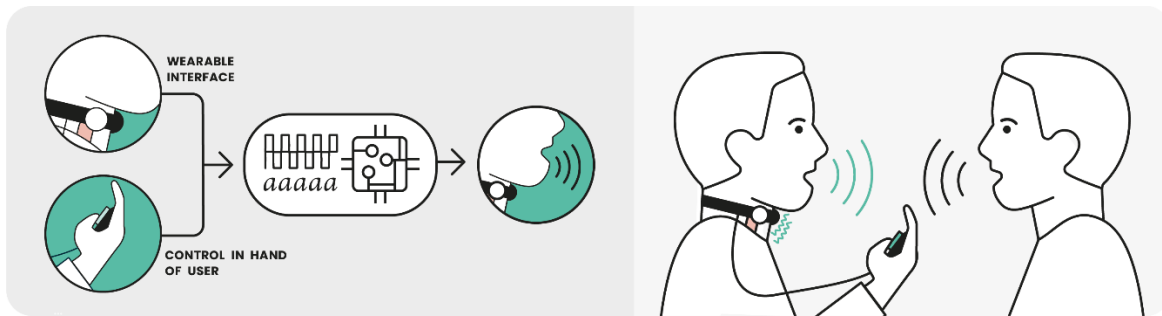


Figure 7: The figure demonstrates the operation of the device by one hand proposed by [24–26]. The person is able to gesture with the hand quite freely, but the control of the onset of the device, and speech characteristics remains (literally) in his or her hand.

mind, they cannot stop the system from producing the “wrong” sentence (causing similar frustration as interfaces predicting a wrong word in a text message). Such a lack of flexibility can be a serious problem and special consideration, as well as evaluation methods should be taken into account while designing this type of system.

3.4 Bionic voices: human-technology close integration/coordination

The term ‘bionic voice’ (coined by [12]), assumes close collaboration and division of labor between the biological and the technical parts of the bionic system, namely a human and a machine performing the respective elements of speech production. Recently this idea was implemented by [1, 2, 20] in Laronix. The system provides voiced air flow to the mouth, while the pressure sensor is located on the stoma, reading the respiration activity of a user. A computational model predicts the adequate vibrations, which are inserted into the mouth by a tube (Figure 6). Voice onset is also controlled by the breath.

This type of system (see also [12] for comparison) requires close human-machine collaboration, imposing constraints on each other, and a human-in-the loop approach is considered (see [12] for definitions). The HCI challenge is to design a system that would enable mastering the device use, which requires constant adaptation to the behavior of the device (adjusting breath) combined with the

coordination with the interlocutors performed at the same time. Based on the recordings available [20], the voice seems natural and, from the coordination point of view, this kind of systems realize an interesting functional division of control: the human controls the intonation and modulation of the voice, and the machine controls the timbre.

3.5 Lean-AI approaches: the physical, and more agentive, direction

We consider lean-AI as an appropriate term for a branch of systems in which the role of artificial intelligence is very limited, leaving maximum relevant control to the human. The system proposed by [35] utilizes a hand controller in the form of a joystick responsible for controlling voice pitch and loudness. The idea of controlling voice onset with a button was already implemented by YourTone company [43], while the original two-dimensional control idea is being developed by the Uhura Technologies [44]. A more advanced use of technology in this type of systems is the signal processing approach to basic waveform modulation [28]. This idea assumes the use of a wearable necklace with an effector producing a more natural-sounding wave (utilizing e.g. a wave constructed from a vowel). From the point of view of coordination, this system puts the laryngectomy in the center of control of many crucial processes in interaction, while providing them with enabling interfaces where

needed (Figure 7). It should be noted, however, that controlling the system with the hand disturbs the interaction by limiting gestures, however the scope of this constraint could be modulated by the design.

4 CONCLUSIONS

On the basis of theoretical foundations of the embodied cognitive science and human-centered computing we have applied the coordination metaphor of communication situated within a particular environment to the analysis of bionic systems for laryngectomees. Specifically, we analyzed constraints that the biological and the technological systems impose on each other, which may enable or disable control needed for proper functioning in interaction. We claim that this framework is suitable for comparative analysis of various forms of speech aid, enabling to assess, besides the intelligibility issues, also the crucial factors for control/agency over interactions.

The novel conceptualization of problems arising from larynx removal surgery, and a problem of “voice restoration” (as we claim: simplified by the literature so far), could be understood as follows: humans are located at the center of the design and evaluation process, and their interactive potential, the ability to coordinate with other people, is supported with technological systems providing the relevant constraints. Five bionic systems have been analyzed in terms of the degree to which they enable or disable control of various aspects of complex coordination of which language is a part. The analyses have allowed us to identify novel concerns for evaluation of the considered systems.

We wish to spark a discussion on conceptualization of speech and human communication in interaction among designers pitching the considerations for, currently neglected, notions of control. The results of these more comprehensive analyses are the new perspectives on the implementation of a continuum of technological solutions. We hope to have shown the usefulness of notions such as coordination, constraints and control in such a debate, beyond the notions of intelligibility and effectiveness of signal transfer. The crucial next step is to design proper evaluation methods to capture this broader set of values meaningful for human interaction.

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